

Scalability & Future Plan

Date	05 July 2024
Team ID	XXXXXX
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited input parameters for crop recommendation (only selected soil and climate factors are considered).	May reduce prediction accuracy under varying agricultural conditions.	High
2	No real-time weather or sensor data integration.	Recommendations may become outdated if environmental conditions change.	High
3	Limited scalability for handling large numbers of users and agricultural datasets.	System performance may decrease as user traffic and data volume increase.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users accessing the system	Deploy cloud-based infrastructure with load balancing and auto-scaling to support more concurrent users.
2	Data Storage	Stores crop, soil, and environmental data in a basic database.	Upgrade to scalable cloud databases with automated backup and efficient data management.
3	Performance	Crop recommendations are generated using existing datasets and algorithms	Optimize machine learning models, implement caching, and improve database query performance for faster recommendations.
4	Security	Basic user authentication and database protection.	Implement secure authentication, role-based access control, data encryption, and regular security monitoring to protect user and agricultural data.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Improve the crop recommendation model by adding more crop varieties and refining prediction accuracy.	1 week	Improves recommendation accuracy and supports more crop options.
Phase 2	Integrate real-time weather data to provide more accurate crop recommendations.	1 week	Enhances decision-making based on current environmental conditions.
Phase 3	Develop a user-friendly interface with multilingual support for farmers.	5 days	Increases accessibility and ease of use for a wider range of users.